

M3 — Live Individual Oral Defense

*Model submission. We are **Group A**. This is the improved analysis and the individual defense of it. Our M1 is in `../M1_replication/`; the critique we received is in `../M2_critique/`; every number below is re-derived in `M3_defense_evidence.qmd`.*

*Read the presentation slides first, then the Q&A. The slides show **what** we did. The Q&A is where it has to hold up.*

Slide 1: What we are defending, and what has moved

Our locked estimand is unchanged: **Group IV vs Group III, weighted Model 1**, on the paper’s own covariate set. What has changed since M1 is how we account for two things around it.

	Source	Outcome
How we describe cohort selection	Reviewer (M2)	Changed — split into two mechanisms
Corroboration from the paper’s Table 2	Reviewer (M2)	Changed — demoted to consistency
The TyG item on “what would change our mind”	Reviewer (M2)	Changed — restated
Which threat dominates	Reviewer (M2)	Retained , with a revised reason
Missing cancer status in Model 1	Ourselves	Tested, no change
Estimand, adjustment set, design, variable definitions	—	Unchanged

Speaker notes: Lead with the traceability table, not with the reviewers. Two of the six rows are ours rather than theirs, and one is a “we tested it and kept it” — the point of M3 is that all three are legitimate.

Slide 2: What the critique got right, and where we push back

Right, and we adopted it. Our M1 cited the paper’s published Table 2 — Group III 3.175 and Group IV 3.155 against Group I, implying about 0.99 — as though it corroborated our locked result. The reviewers are correct that this is not independent confirmation: it is the same data, and our own detail note already conceded no interval is recoverable because the covariance of the two published coefficients is not reported. **We have demoted it to a consistency observation.**

Right, and we adopted it. Our “what would change our mind” list included a design-aware re-run of the authors’ TyG sensitivity. The reviewers are right that this could not settle the locked question whatever weights were used, because that analysis is against Group I. **We have restated the item** to say what a TyG re-analysis would actually have to estimate.

Partly right — and it forced the most useful change we made. They nominate selection into the cohort as the dominant threat. We accept selection is consequential and that our M1 under-described it. But their slide treats it as **one** mechanism, and it is two. See Slide 3.

Where we push back. They rank selection above estimand–adjustment-set alignment on the grounds that an unstated estimand is “a repairable reporting failure.” We think that understates it: if the target quantity is not named, there is no fact of the matter about whether the adjustment set is right, so the alignment problem is *prior* to the selection problem rather than smaller than it. We retain our ranking, on a narrower argument than we gave at M1 — see Slide 5.

Slide 3: Reviewer-prompted work — the selection is two mechanisms, not one

The evidence pack counts them separately:

	n
adults 18+	36,580
(1) sampled into the fasting subsample	14,962
(2) of those, FLI computable	14,170
lost at step (2)	792

- **(1) is selection by design.** NHANES decided who was asked to fast. It is known and planned, and WTSAF2YR exists to account for it — which is why we weight by it and subset the design rather than the rows. Under the survey design this step is *handled*.
- **(2) is item-level availability.** Among people already sampled to fast, FLI still needs triglycerides, GGT, BMI and waist all present. No design weight accounts for that, and whether it is ignorable depends on why those measurements are absent — which the public files do not tell us.

They bound different things. (1) bears on the **target population**, and the weights handle it: WTSAF2YR is built so that the fasting subsample represents the US civilian non-institutionalized population, so our weighted estimate speaks to US adults meeting the MASLD criteria, not only to those sampled to fast. (2) is a potential **selection bias within** that population, and it is the one that could distort the contrast itself.

Speaker notes: This is the change we are proudest of and it came from the review. Be explicit that we are not conceding the ranking by adopting the distinction — adopting a better description of a threat is not the same as agreeing it dominates.

Slide 4: Self-initiated work — could incomplete cancer status explain the result?

No reviewer asked for this. Cancer status is the only Model-1 covariate carrying any missingness, and we were not willing to defend an estimate that rests on dropping people without testing it. On the locked contrast, design-aware, with the paper's Model 1 unchanged; complete case uses the 5,939 rows complete on Model 1, and the other three keep all 6,048 rows of the locked domain:

Locked estimand (IV vs III), weighted Model 1	HR (95% CI)
complete case	1.29 (0.89–1.86)
all missing assigned no cancer	1.28 (0.89–1.85)
all missing assigned cancer	1.28 (0.89–1.85)
MI (m = 10), outcome in the imputation model	1.28 (0.89–1.85)

- **The four are materially indistinguishable.** Complete case gives 1.29; both extreme assignments and MI give 1.28, a difference of 0.01, far smaller than the interval. That is the finding, and it is a result: the estimate we defend does not depend in any meaningful way on how the incomplete records are handled.
- **We did not widen the adjustment set to produce a comparison.** Adding covariates until the missingness bites would have changed the estimand and answered a question nobody asked.
- **The two extreme assignments are transparent global scenarios, not bounds.**

What this does not do. It says nothing about selection. It concerns incomplete cancer status among people **already in the cohort**; it cannot speak to anyone who never entered it. Stability under missing-data sensitivity is not transportability, and we are not offering it as an answer to Slide 3.

Speaker notes: The last paragraph is the one to say slowly. The tempting move is to let “the estimate held” absorb the reviewers’ selection objection, and it does not touch it.

Slide 5: The analysis we defend

- **Estimand, adjustment set, survey design and variable definitions: unchanged from M1.** Nothing in the critique or in our own re-analysis gave us a reason to move them, and changing them would have made M1 and M3 incomparable.
- **The locked estimate stands at 1.29 (0.89–1.86):** Group IV vs III, the paper’s Model 1, design-weighted, complete case on 5,939 rows. The three sensitivity analyses give 1.28 (0.89–1.85).
- **Dominant threat: we retain estimand–adjustment-set alignment,** on a narrower argument than M1 gave. Not “the estimand is unstated and that is bad”, but: *until the target quantity is named, no adjustment set can be shown to be the right one, and the attenuation has no unique interpretation.* Selection then bounds where the answer travels — which is why we now report the two mechanisms separately rather than folding them into one bullet.
- **What would still change our mind:** a stated target quantity with a causal model supporting the adjustment set; evidence on why FLI components are missing at step (2); or a pooled sample large enough to resolve the interaction.

Speaker notes: Close on what is unchanged, not on what moved. A defense that lists only changes implies the original was mostly wrong, and it was not.

Q&A Transcript

Examiner: Your headline number moves by a hundredth at most across four missing-data approaches. Is that reassuring, or does it just mean the sensitivity analysis had nothing to work with?

Student (AB): Both, and the second is the honest half. Only 109 observations are incomplete, and only on one covariate, so no reasonable handling of them could move a hazard ratio far. What the analysis rules out is a specific alternative explanation — that our estimate is an artefact of dropping those records. It does not demonstrate robustness in general. If someone wants to know whether the estimate is fragile, missing cancer status is the wrong place to look; selection at step (2) is the right one, and we cannot test it with these data.

Examiner: Your reviewers said selection dominates. You kept your own ranking. Convince me that is not just defending your original position.

Student (CD): We changed our description of selection because they were right about it, and we changed two other things they asked for. Where we differ is on ordering. Their argument is that an unstated estimand is repairable in a sentence. Ours is that the sentence is not cosmetic: without a named target quantity, “is this adjustment set correct?” has no answer, so we cannot even say whether the attenuation reflects confounding control or over-adjustment. Selection tells you where an interpretable estimate travels. Alignment tells you whether you have an interpretable estimate to travel with. We think the second comes first. We would have conceded if they had shown that naming the estimand leaves the analysis unchanged — that is the argument that would move us.

Examiner: You ran a missing-data analysis nobody asked for. Why was that a good use of your time rather than a detour?

Student (EF): Because we would have been asked today, and “we did not check” is a worse answer than any number we could have got. It also disciplined the rest of the work: once we had decided to test the locked contrast rather than a convenient wider model, it was obvious that the widening was the thing to avoid. An earlier draft of ours added income and systolic blood pressure so the comparison would show movement. That would have been a different adjustment set and a different estimand, and it would have manufactured a finding rather than tested one. We deleted it.

Examiner: Which part of this was yours?

Student (EF): The missing-data work on Slide 4, including the decision to keep the outcome in the imputation model rather than imputing on covariates alone, and the decision to drop the wider model. CD

wrote the response on ranking; AB re-ran the design-aware fits and checked that the domain is subsetting rather than the rows.

Examiner: If you had one more week?

Student (AB): Step (2). We would try to characterise who loses an FLI component and why — whether it tracks anything measured — because that is the mechanism that could bias the contrast rather than just relocate the population. Everything else we would leave alone.